Rongwu Xu

Personal

Pronouns:	He/Him/His
Location:	FIT building, Tsinghua University, Beijing, China
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Research

I delve into the *intersection* of natural language processing and trustworthy AI, with a focus on safety and ethics issues in large language models (LLMs). My concentration encompasses two primary areas:

- **Investigating and Enhancing Trust in LLMs**. I investigate and address the inherent safety, ethics, and bias concerns present in LLMs. My ultimate goal is to develop strategies that mitigate these issues, contributing to the creation of more trustworthy AI systems that align with human values.
- **Utilizing NLP in Public Discourse and Moderation**. I apply the advancements in NLP to analyze public opinions and to enhance content moderation. This involves leveraging cutting-edge language techniques to understand, interpret, and manage the vast array of information and viewpoints expressed in digital platforms.

My vision is to harness the power of AI and NLP to create safe, ethical, and human-centric digital environments.

Education

06/2025 (EST.) — MRes in Computer Science
Tsinghua University
Beijing, China
Advisor: Prof. Wei Xu
08/2022 — Graduate Student at IIIS (Headed by Andrew C. Yao, the Turing award laureate'2000)

06/2022 — BEng in Computer Science
Tsinghua University
Beijing, China
09/2018 — Bachelor Student at Department of Computer Science and Technology (DCST)

Publications and Manuscripts

11. Preemptive Answer "Attacks" on Chain-of-Thought Reasoning
Rongwu Xu*, Zehan Qi*, Wei Xu
Preprint

10. Understandable and Singable Musical Lyrics Translation
Jinhan Li, Zhuorui Ye, **Rongwu Xu**
Preprint

9. Perspective-taking Gives More Ethical Responses
Rongwu Xu, Zi'an Zhou, Tianwei Zhang, Zehan Qi, Su Yao, Ke Xu, Wei Xu, Han Qiu
Preprint

8. Knowledge Conflicts for LLMs: A Survey
Rongwu Xu*, Zehan Qi*, Cunxiang Wang, Hongru Wang, Yue Zhang, Wei Xu
arXiv Preprint

7. The Earth is Flat because...: Investigating LLMs' Belief towards Misinformation via Persuasive Conversation
Rongwu Xu, Brian S. Lin, Shujian Yang, Tianqi Zhang, Weiyan Shi, Tianwei Zhang, Zhixuan Fang, Wei Xu, Han Qiu
arXiv Preprint

6. Exploring Chinese Humor Generation: A Study on Two-part Allegorical Sayings
Rongwu Xu
In *International Joint Conference on Neural Networks* (**IJCNN**), 2024

5. Tempo: Confidentiality Preservation in Cloud-based Neural Network Training
Rongwu Xu and Zhixuan Fang
In *International Joint Conference on Neural Networks* (**IJCNN**), 2024

4. LSync: A Universal Timeline-synchronizing Solution for Live Streaming
Yifan Xu*, Fan Dang*, **Rongwu Xu**, Xinlei Chen, Yunhao Liu
In *IEEE/ACM Transactions on Networking* (**ToN**), 2024

3. MISO: Legacy-compatible Privacy-preserving Single Sign-on using Trusted Execution Environments
Rongwu Xu, Sen Yang, Fan Zhang, Zhixuan Fang
In *IEEE European Symposium on Security and Privacy* (**EuroS&P**), 2023

2. LSync: A Universal Event-synchronizing Solution for Live Streaming
Yifan Xu, Fan Dang, **Rongwu Xu**, Xinlei Chen, Yunhao Liu
In *IEEE Conference on Computer Communications* (**INFOCOM**), 2022

1. LifeRec: A Mobile App for Lifelog Recording and Ubiquitous Recommendation
Jiayu Li, Hantian Zhang*, Zhiyu He*, **Rongwu Xu***, Pingfei Wu*, Min Zhang, Yiqun Liu, Shaoping Ma
In *ACM SIGIR Conference on Human Information Interaction and Retrieval* (**CHIIR**), 2022

(* equal contribution)

Awards

2023	Overall Excellence Scholarship	Tsinghua University
2020	Technological Innovation Scholarship	Tsinghua University
2020	First Prize in "Youth in Action" Social Practice	Tsinghua University
2019	Tsinghua-Panasonic Scholarship	Tsinghua University

Talk and Presentation

May. 2023	Privacy-preserving authentication	Oral report@EuroS&P conference

Experience

Mentoring

Dec. 2023 -	Zi'an Zhou	Undergrad, Zhili College (Information and Computational Science)@Tsinghua Univ.
Sep. 2023 - Feb. 2024	Shujian Yang	Master, SPEIT (Information Engineering)@SJTU
Jun. 2023 -	Tianqi Zhang	Undergrad, CS@Tsinghua Univ.
Jun. 2023 - Jan. 2024	Brian S. Lin	Undergrad, CS@Tsinghua Univ.
Oct. 2021 - Jul. 2022	Xingyu Dang	Undergrad, IIIS@Tsinghua Univ.

Teaching

Spring 2024 **Teaching Assistant & Organizer**, Tsinghua University
Introduction of Large Language Model Applications
Co-designed labs, organized the curriculum and assisted labs in class.

Fall 2023 **Teaching Assistant**, Tsinghua University
Operating System and Distributed System
Held discussion and office hours, graded exams, assignments and projects.

Spring 2022 **Teaching Assistant**, Tsinghua University
Distributed System and Blockchain
Held discussion and office hours, graded exams, assignments and projects.

Exchanging

Apr. 2021 - Oct. 2022 **Research Assistant** (Remote), Duke University
Granted summer overseas internship (undergrad) by Tsinghua
Conducted research in privacy-preserving authentication.
Host: Prof. Fan Zhang

Internship

Dec. 2022 - Jan. 2023 **Shanghai Qi Zhi Institute**, Shanghai, China
Conducted research in decentralized finance (DeFi). Worked on MEV arbitrage forecasting algorithms using graph neural networks (GNNs).
Mentor: Prof. Zhixuan Fang

Skills and Expertise

- **Research:** Experienced in deep learning/data analysis, also ability with computer systems/applied cryptography/software engineering.
- **Programming Language:** Proficient in C++/Go/Python/Java/JavaScript/LaTeX, also ability with Verilog/System Verilog/ASM/Rust/P4/HTML/Matlab.
- **Technological:** Proficient in Pytorch/NumPy/Matplotlib/Git/Linux OS/Markdown/, also ability with Docker/Wireshark/Microsoft Office.
- **Other Expertise:** Good communication skills, love collaborating with people, experienced in mentoring students, works well in a team.

Services

Academic

Mar. 2024 - Current **Member**, ACL SIGSEC, Association for Computational Linguistics

Societal

Nov. 2023 - Current **Member**, IIIS Students Congress, Tsinghua University

— Last updated on Saturday 16th March, 2024 —